

CHM102 Basic Organic Chemistry

Spring semester

January 2026

Instructor: Dr Sugumar Venkataramani,
Office: 5F10, AB2

Tutors: Dr Archana Velloth & Dr. Akanksha Kumari

Lecture timing Venue: LH6 Wednesdays 3.00 to 4.00 pm Thursdays 3.00 to 4.00 pm Fridays 3.00 to 4.00 pm		Schedule						
		JANUARY 2026						
		SUN	MON	TUE	WED	THU	FRI	SAT
						1 Registration for new PhD students	2	3 Last date of registration for all courses
		4	5 Announcement of course cancellations Last date of course cancellations with LHM PhD Co-ordinator of course registrations	6	7	8	9	10
I Mid-sem exams February 4, 5, 6 & 7, 2026		11	12	13 Qualifier: adding course to replace a dropped course (no new registrations)	14	15	16	17
		18	19	20	21 Last date for submission of annual reports and annual review for PhD students	22	23	24
		25	26 Republic Day	27	28	29	30	31
II Mid-sem exams March 3, 5, 6 & 7, 2026		FEBRUARY 2026						
		SUN	MON	TUE	WED	THU	FRI	SAT
		1	2	3	4 1st Mid Sem Exams (only for preceptors)	5 1st Mid Sem Exams (all batches)	6 1st Mid Sem Exams (all batches)	7 1st Mid Sem Exams (all batches)
		8	9	10	11	12	13	14
		15	16 (BSMS Students and Mentors Meeting week)	17	18 PhD & MPhil PhD advertisement appears	19	20	21 Buffer day for teaching & rescheduled class (if any)
End-sem exams April 27 – May 7, 2026		22	23	24	25	26	27	28 Science Day
Total – 38 lectures {Before I midsem: 13 lectures; Between I & II mid-sem: 8 lectures; After II mid-sem: 17 lectures}								

Schedule

Lecture timing

Venue: LH6

Wednesdays 3.00 to 4.00 pm

Thursdays 3.00 to 4.00 pm

Fridays 3.00 to 4.00 pm

I Mid-sem exams

February 4, 5, 6 & 7, 2026

II Mid-sem exams

March 3, 5, 6 & 7, 2026

End-sem exams

April 27 – May 7, 2026

Total – 38 lectures

{Before I midsem: 13 lectures; Between I & II mid-sem: 8 lectures; After II mid-sem: 17 lectures}

MARCH 2026						
SUN	MON	TUE	WED	THU	FRI	SAT
1	2	3 2nd Mid Sem Exams (only for pre-majors)	4 National Safety week (6-28 March 2024)	5 2nd Mid Sem Exams (all batches)	6 2nd Mid Sem Exams (all batches)	7 2nd Mid Sem Exams (all batches)
8 Mid Term Break	9 Mid Term Break	10 Mid Term Break	11 Mid Term Break	12 Mid Term Break	13 Mid Term Break	14 Mid Term Break
15 Mid Term Break	16	17	18	19	20	21 Mid Ftr
22	23	24	25	26	27 PhD & int. PhD Registration deadline Finalization of teaching arrangements for Munich semester	28 Buffer day for teaching a rescheduled class (if any)
29	30 Form 1: MS thesis intent form submission deadline	31 Mahavir Jayanti				
APRIL 2026						
SUN	MON	TUE	WED	THU	FRI	SAT
			1 Pre-majors open house	2 Majors and Minors open house	3 Good Friday	4
5	6 Feedback Collection Meet Dept Chair, MS Thesis with copy of MS and int. PhD MS thesis submission to the committee deadline	7	8	9	10 Pre-registration for the Summer Semester 2026, PhD and others for interviews sent out	11 MS thesis open seminar / poster session
12	13	14	15	16	17	18 Buffer day for teaching a rescheduled class (if any)
19	20 Lab Exams and Theory classes as per the schedule	21 Lab Exams and Theory classes as per the schedule	22 Lab Exams and Theory Classes as per the scheduling	23 Lab Exams and Theory classes as per the schedule	24 Lab Exams and Theory classes as per the schedule	25 Lab Exams and Theory classes as per the schedule
26	27 End Sem Exams (all batches)	28 End Sem Exams (all batches)	29 End Sem Exams (all batches)	30 End Sem Exams (all batches)		

Grading Policy

	Plan A	Plan B
I – Mid-sem exam	20	15
II – Mid-sem exam	20	15
End-sem exam	50	50
Assignments	10	5
Surprise quizzes/ Attendance	0	15

***Plan B will be activated if the attendance drops significantly**

CHM102: Basic organic chemistry [Cr:3, Lc:2, Tt:1, Lb:0]
Course Outline

- A basic introduction to the importance of organic molecules in life.
- Nomenclature of organic compounds with diverse structure and functional groups.
 - Concepts involving acidity, basicity, tautomerism, inductive, mesomeric, Nathan field and steric effects.
- Stereochemistry: Conformational analysis of acyclic (ethane, butane, substituted acyclic compounds) and cyclic (cyclohexane, cyclopentane) systems. Energies and conformational preferences of differently substituted cyclohexanes and related systems.
 - Axial (atropisomerism) and planar (helical) chirality. Optical isomerism in biphenyls, allenes.
 - Geometrical isomerism (E/Z), acyclic and cyclic compounds.
 - Description of syn/anti (erythro, threo), meso isomers. Absolute and relative stereochemistry.
 - R/S configuration assignment. Racemization, resolution, prostereoisomerism, stereotopicity and enantiomeric excess, d,l isomers and racemate.
- Reaction types and basic reaction mechanisms involving:
 - Substitution reactions: Nucleophilic substitution at carbon and its mechanism (including chirality generation). Grignard types reagents and their addition to carbonyls.
 - Addition reactions: Addition of X, HX, boranes and hydroxylation to C=C systems and epoxidation of olefins. Stereoaspects of the addition of X, HX, X₂, boranes and hydroxylation to C=C systems. Cis- and trans-hydroxylation of cycloalkenes and related reactions.
 - Elimination reactions: Elimination reactions leading to C=C bond formation and their mechanisms including named reactions (e.g., Hoffman and Saytzeff eliminations).
- Selectivity in organic reactions: Chemoselectivity, regioselectivity, enantio- and stereo-selectivity (explanation with examples).
- Reactive intermediates: Carbenes, nitrenes, carbocations, carbanions, and radicals. Reactions involving reactive intermediates, rearrangement reactions and mechanisms.
- A basic introduction to thermodynamic vs kinetic control, Hammond postulate (energy profile diagram).
- Aromaticity and anti-aromaticity: Hückel and Craig's rules, homo, hetero (furan, thiophene and pyrrole) and non-benzenoid aromatic systems. Polyaromatic compounds.

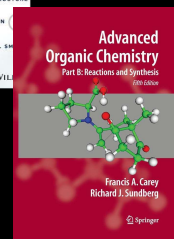
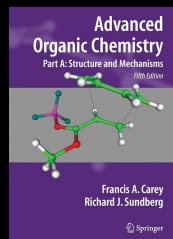
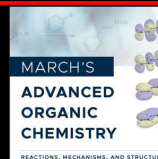
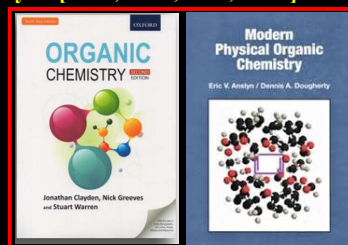
CHM102: Basic organic chemistry [Cr:3, Lc:2, Tt:1, Lb:0]

Recommended Reading

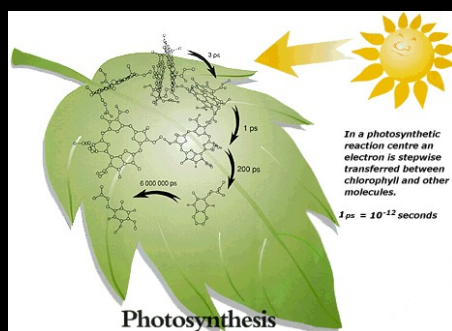
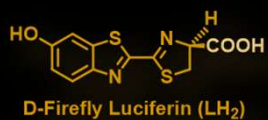
- J. Clayden, N. Greeves, S. G. Warren, *Organic Chemistry*, 2nd Ed, Oxford University Press, Oxford (2012).
- R. T. Morrison, R. N. Boyd, S. Bhattacharjee, *Organic Chemistry*, Indian Edition, 6th Ed, Pearson Education, New Delhi (2007).
- E. V. Anslyn, D. A. Dougherty, *Modern Physical Organic Chemistry*, University Science Books, 2005.

Suggested Reading

- M. B. Smith, *March's Advanced Organic Chemistry: Reactions, Mechanisms, and Structure*, 8th Ed, John Wiley & Sons, New Jersey (2020).
- P. Sykes, *A Guidebook to Mechanism in Organic Chemistry*, Indian Edition, 7th Ed, Pearson Education, New Delhi (2010) (Dorling Kindersley (India) Pvt Ltd).

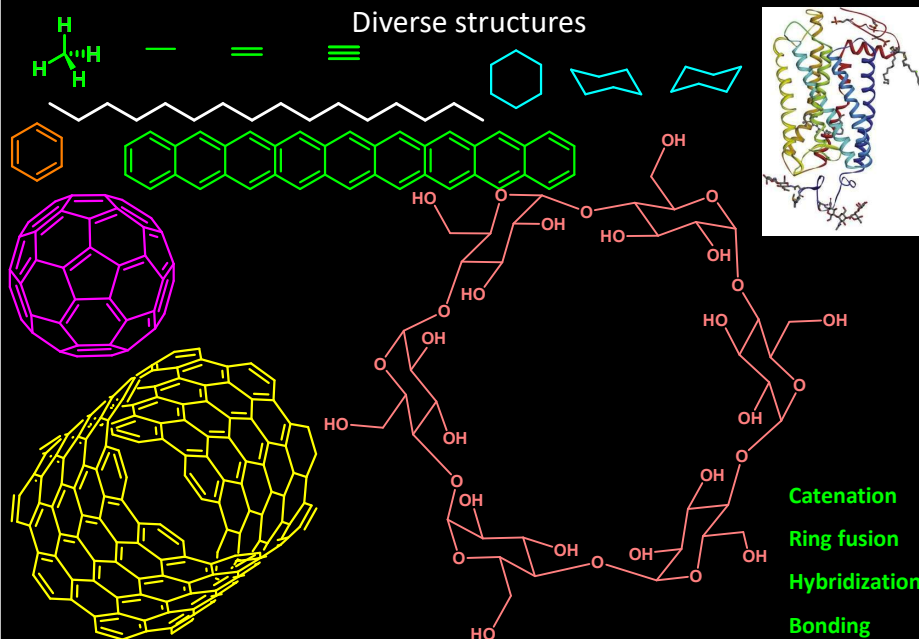


Why organic chemistry?

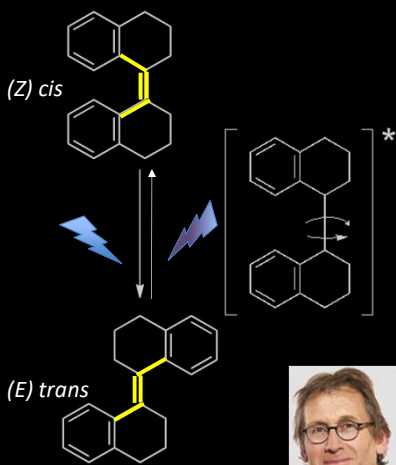


Why organic chemistry?

Diverse structures



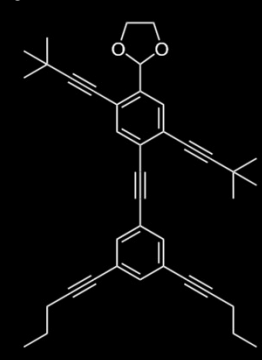
Why organic chemistry?
Fantasy world



(Z) cis

(E) trans

Nature, 1999, 401, 152



NanoPutian

2-(2,5-bis(3,3-dimethylbut-1-yn-1-yl)-4-[[3,5-di(pent-1-yn-1-yl)phenyl]ethynyl]phenyl)-1,3-dioxolane

Nomenclature
Isomerism
Photochemistry

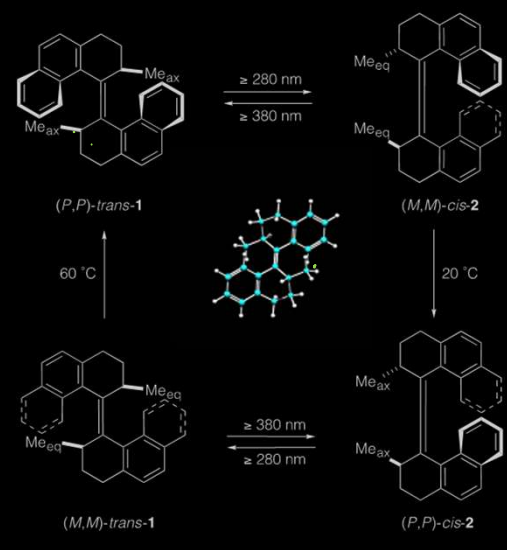


B. L. Feringa

Why organic chemistry?
Fantasy world
Light-driven monodirectional molecular rotor



Nature, 1999, 401, 152



(P,P)-trans-1

(M,M)-cis-2

(M,M)-trans-1

(P,P)-cis-2

60 °C

20 °C

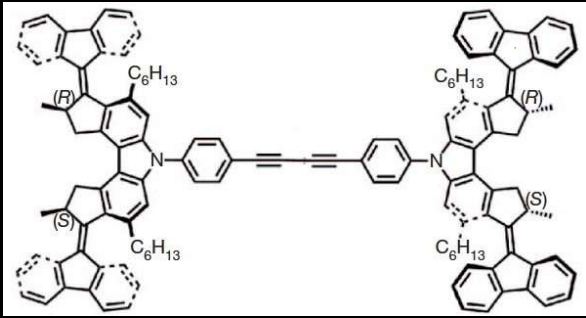
$\approx 280 \text{ nm}$

$\approx 380 \text{ nm}$

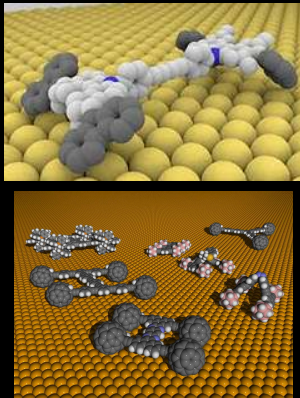


B. L. Feringa

Why organic chemistry?
Fantasy world



The "real" Nano Car



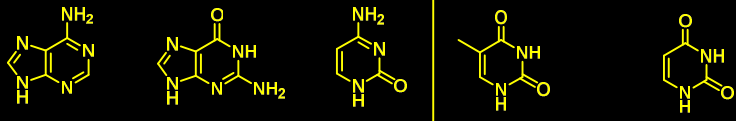


B. L. Feringa



Why organic chemistry?
Nature's choice

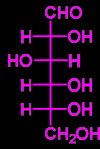
Nucleobases



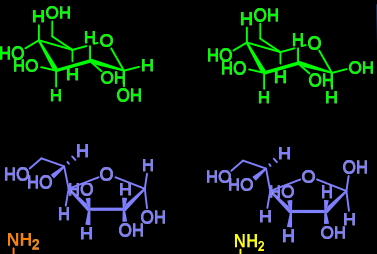
Adenine Guanine Cytosine Thymine Uracil

+

Carbohydrates

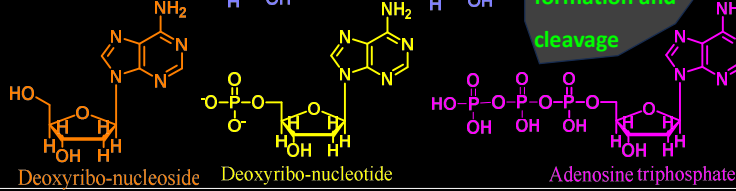


D-Glucose



↓

Nucleic acids
DNA and RNA



Deoxyribo-nucleoside Deoxyribo-nucleotide Adenosine triphosphate

Heterocycles

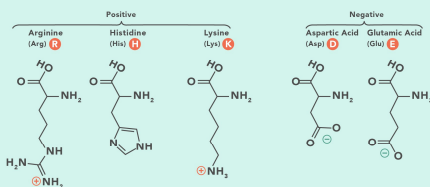
Conformations

Ether linkage

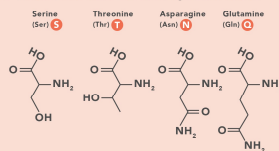
Energy storage and release by bond formation and cleavage

Why organic chemistry? Nature's choice

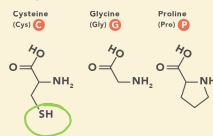
A. Amino Acids with Electrically Charged Side Chains



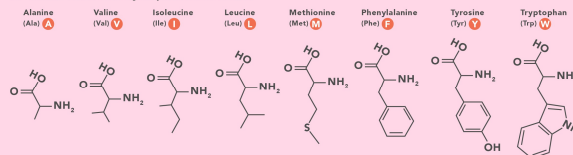
B. Amino Acids with Polar Uncharged Side Chains



C. Special Cases



D. Amino Acids with Hydrophobic Side Chains



Amino acids

↓
Peptides

↓
Proteins

↓
Structure and functions

Functional groups

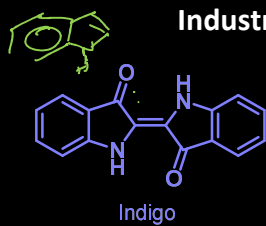
Amide bonds

Hydrogen bonding

Catalysis

Why organic chemistry?

Industrial relevance – Dyes and pigments

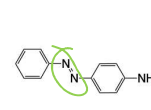
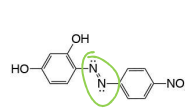
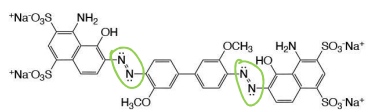
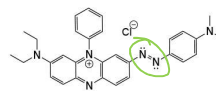
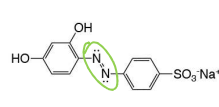
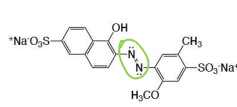
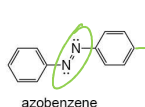


Resonance

Extended conjugation

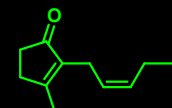
Light absorption

Chromophores

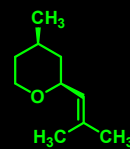
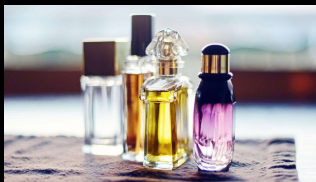


Why organic chemistry?

Industrial relevance – Perfumes and Flavoring agents



cis Jasmone
jasmine perfume



(-)-cis Rose oxide
Rose oil



Alkyl pyrazine
From coffee & roasted meat



Corylone
Caramel roasted taste

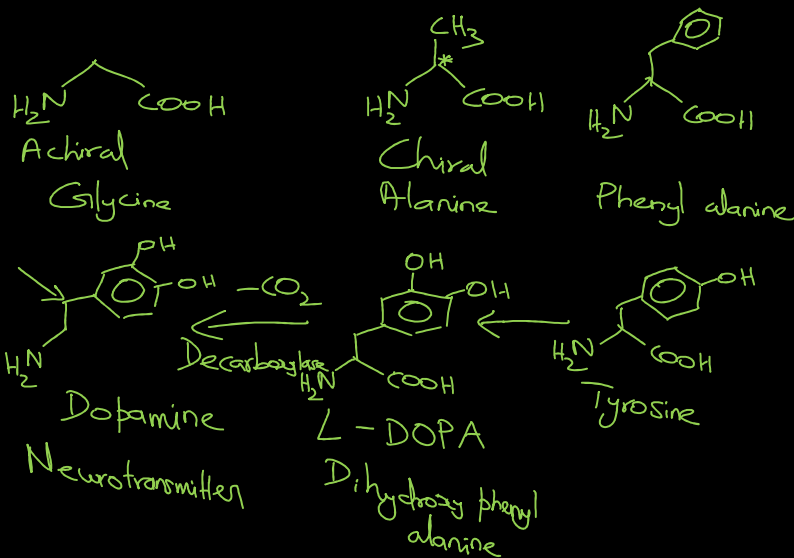


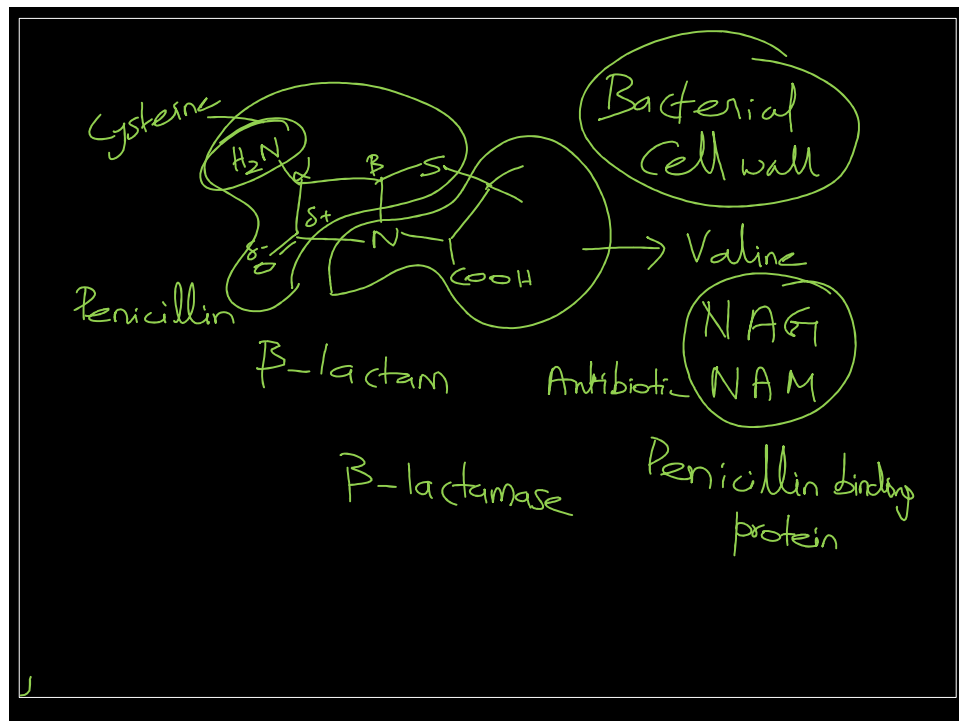
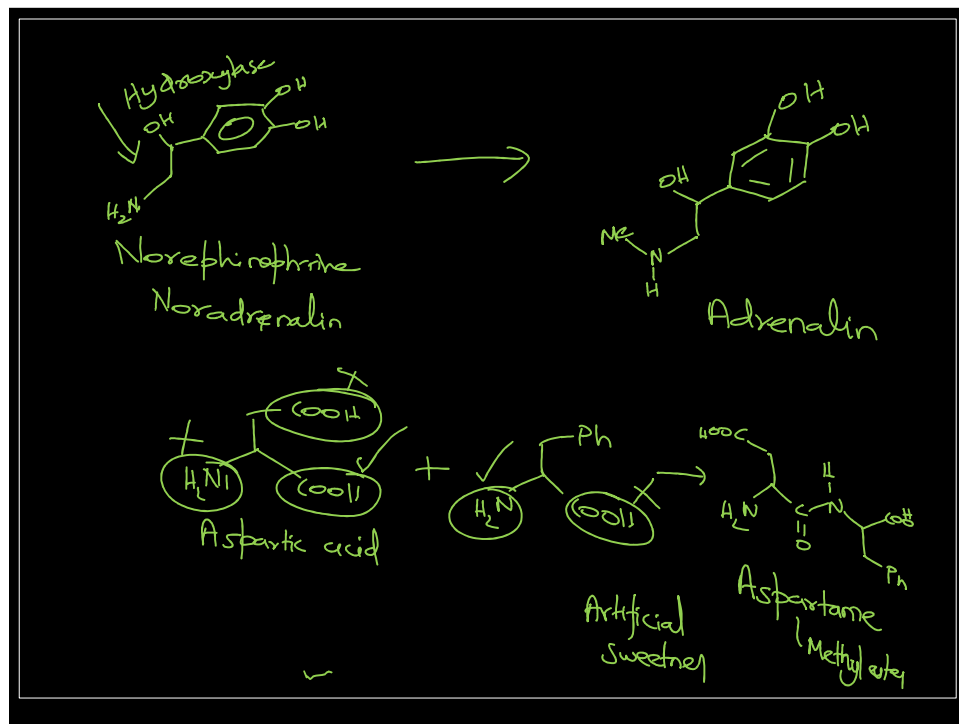
Vanillin



CHM102: Basic organic chemistry

- A basic introduction to the **importance of organic molecules in life.**



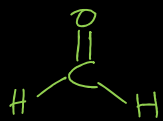


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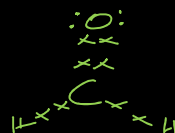
CHM102: Basic organic chemistry

- Nomenclature of organic compounds with diverse structure and functional groups.

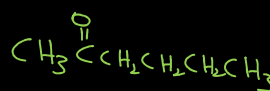
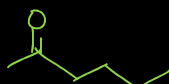
Why nomenclature is Important?



Kekule structure
Line drawing



Lewis structure



Condensed structures

CH_4 Methane

CH_3CH_3 — Ethane

$\text{CH}_3\text{CH}_2\text{CH}_3$ Propane

$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ Butane

$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ Pentane

$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ Hexane

$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ Heptane

$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ Octane

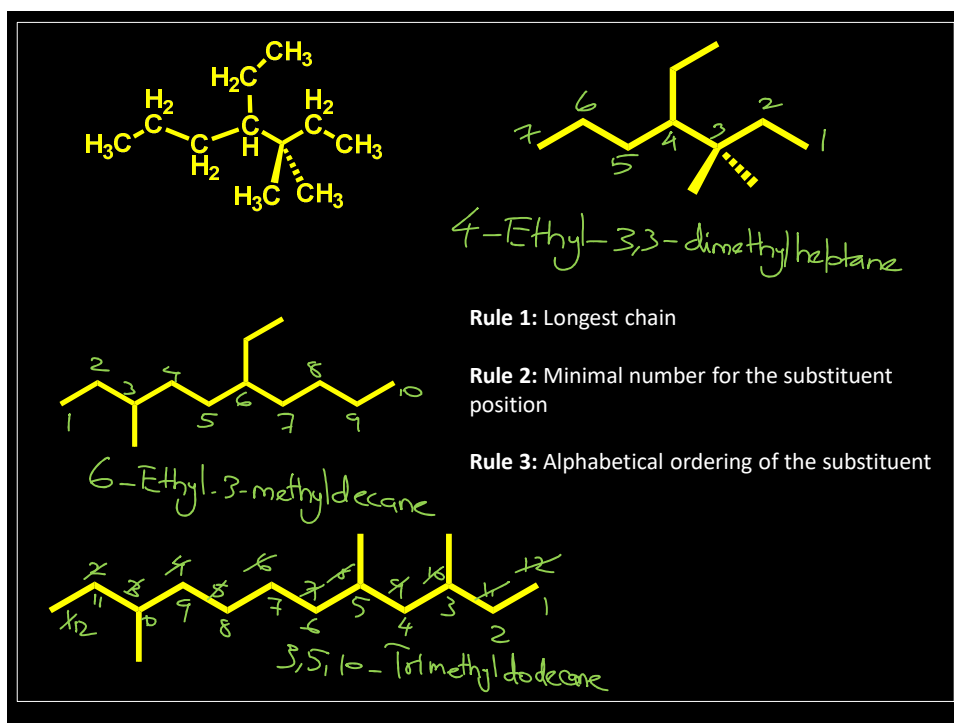
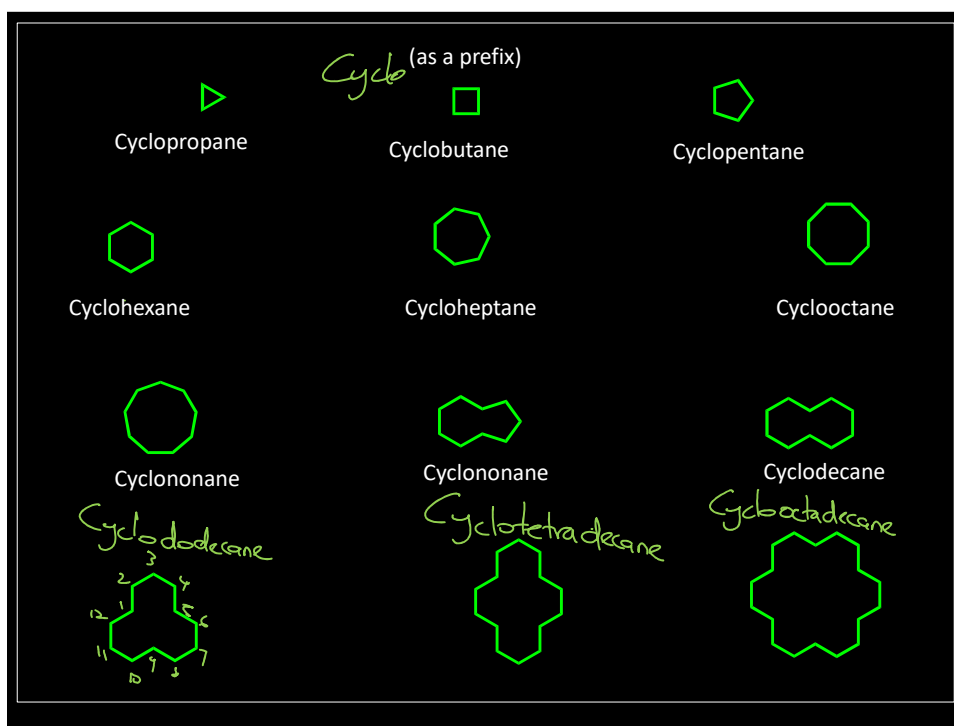
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ Nonane

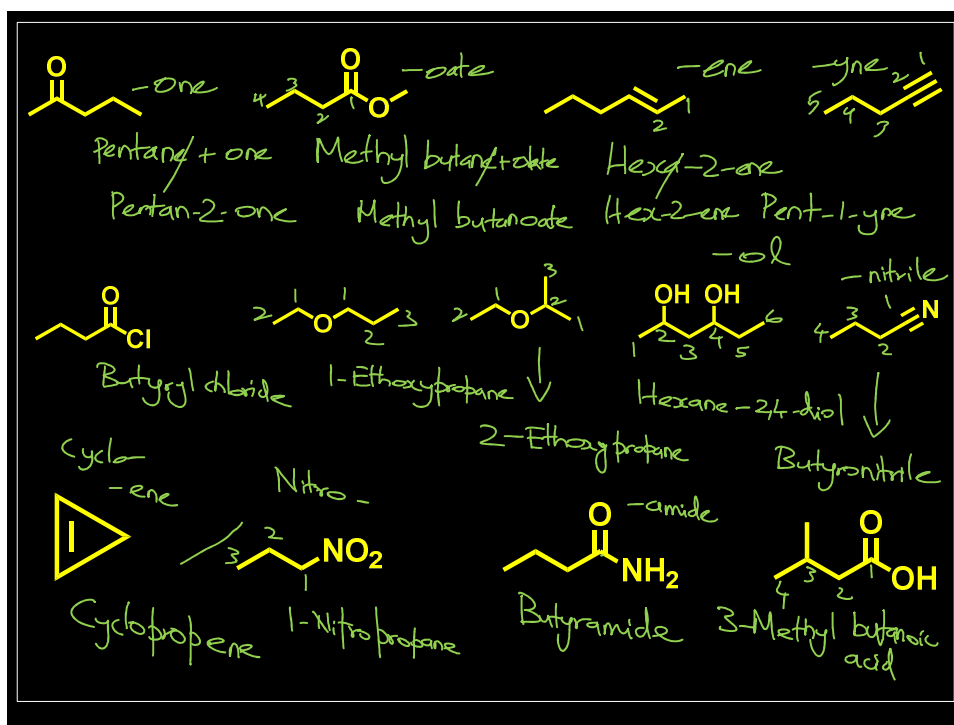
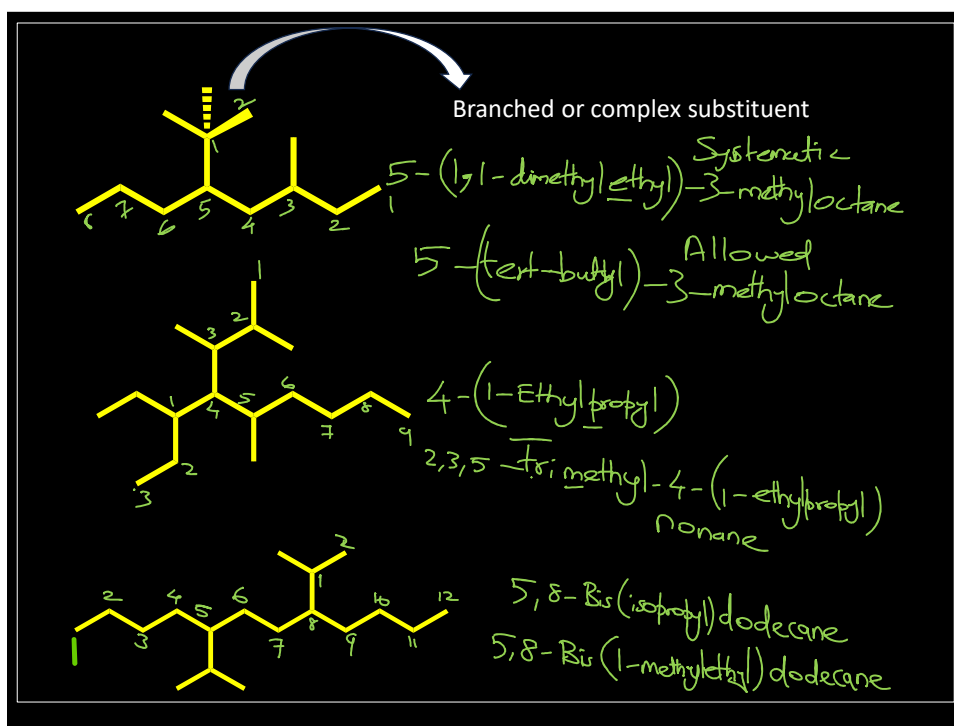
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ Decane

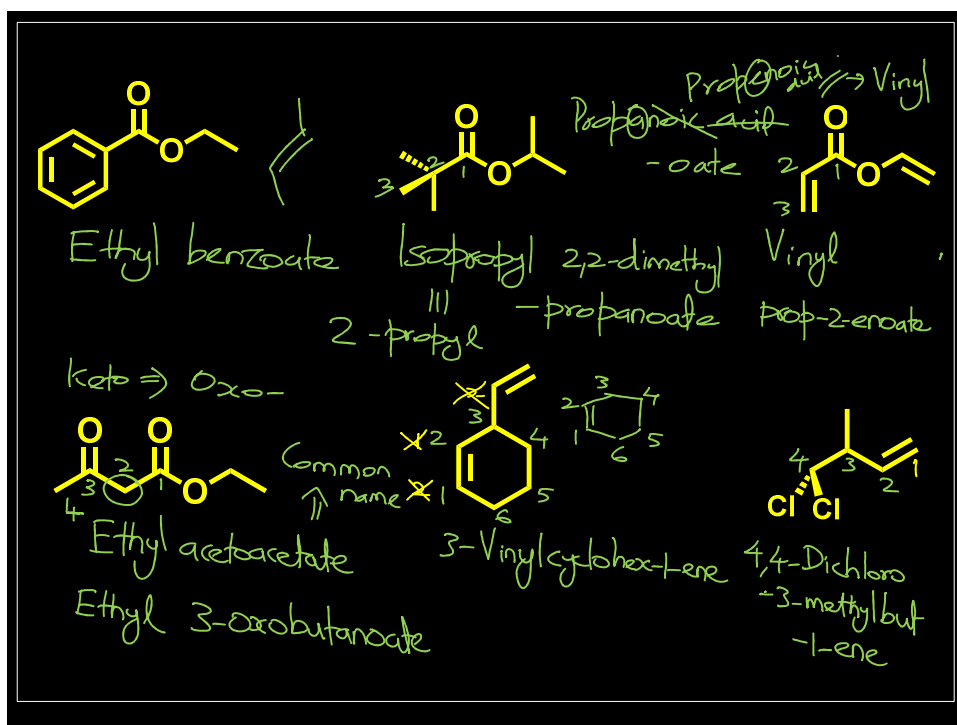
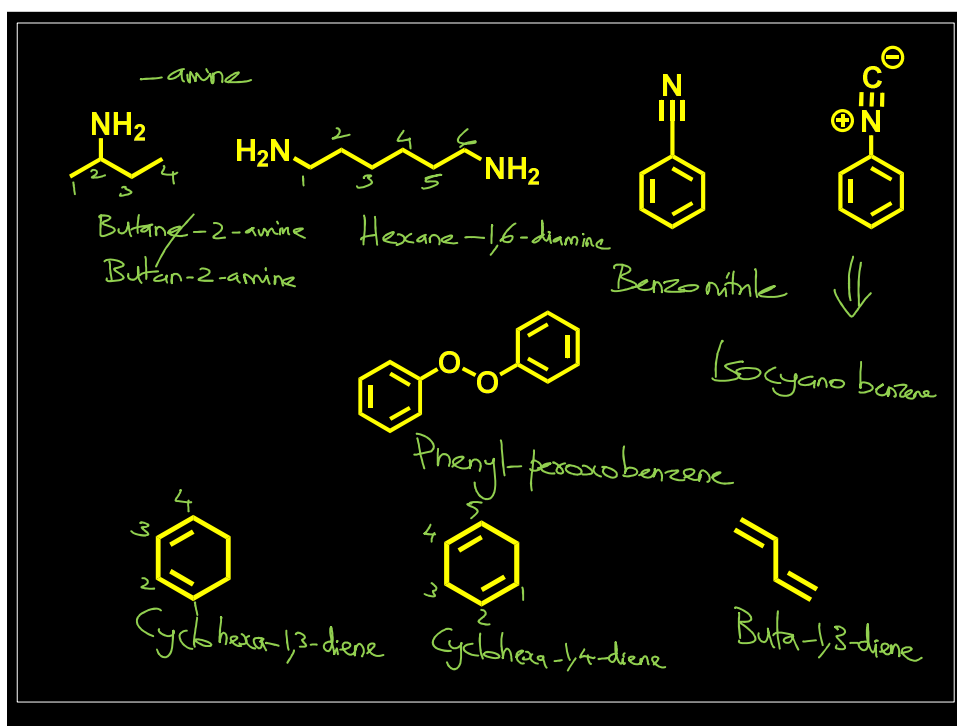
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ Undecane

$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ Dodecane

C_{18} - Octadecane







Nomenclature – IUPAC rules

- Identify the longest carbon chain containing the functional group with the highest priority. This chain determines the *stem* or *parent name* of the compound.
- Identify the substituents attached to the long alkyl chain
- Numbering of the long alkyl chain begins at the end closest to the substituent(s)
- Alphabetic ordering of substituents (e.g. ethyl > 2,3-dimethyl)
- Systematic names of complex substituents should be enclosed in parenthesis **()**
- If complex substituent appears multiple times, prefixes (bis, tris, tetrakis, pentakis) should be used.
- Change the ending of the parent alkane/alkene/alkyne to the *suffix* of the highest priority group, which gives the parent name of the compound (usually, drop the last letter "e" before adding the suffix, if the suffix starts with vowel).
- Number the chain from the end closest to the highest functional group.
- The other groups are named as substituents by using the appropriate *prefixes*.
- Assign stereochemistry, *E/Z* or *R/S*, as necessary

Table 3: Seniority order for characteristic groups.

Class	Formula*	Suffix	Prefix
Carboxylates	–COO [–]	carboxylate	carboxylato
	–(C)OO [–]	oate	
Carboxylic acids	–COOH	carboxylic acid	carboxy
	–(C)OOH	oic acid	
Esters	–COOR	(R) ...carboxylate**	(R)oxycarbonyl
	–(C)OOR	(R) ...oate**	
Acid halides	–COX	carbonyl halide	halocarbonyl
	–(C)OX	oyl halide	
Amides	–CONH ₂	carboxamide	carbamoyl
	–(C)ONH ₂	amide	
Nitriles	–C≡N	carbonitrile	cyano
	–(C)≡N	nitrile	
Aldehydes	–CHO	carbaldehyde	formyl
	–(C)HO	al	oxo
Ketones	=O	one	oxo
Alcohols	–OH	ol	hydroxy
Thiols	–SH	thiol	sulfanyl***
Amines	–NH ₂	amine	amino
Imines	=NH	imine	imino

* Here –(C) indicates that the carbon atom is implied by the parent name.

** Here (R) means that the group R is expressed as a separate prefixed word.

*** Note: 'mercapto' is no longer acceptable (but is still used by CAS).

Pure Appl. Chem. 2020; 92(3): 527–539

Only Suffix		Only Prefix	
Olefin	-ene	Halogen	halo-
Alkyne	-yne	NO ₂	Nitro-

